SAT Math Final Revision Set 2

76 questions · 2 hours

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (13 items)

1. A-317 · Hard · Multiple choice · **No calculator needed**

The function f is linear, $f(5) + f(-5) = 14$, and $f(2) = 13$. What is the value of $f(-2)$?

A) -13

B) 1

C) -1

D) 7

2. A-320 · Hard · Multiple choice · **No calculator needed**

In the equation $ax + by = c$, a, b and c are constants with $a > 0$, $b < 0$ and $c > 0$. The graph of the equation in the xy-plane does NOT pass through which quadrant?

A) Quadrant II

B) Quadrant III

C) Quadrant I

D) Quadrant IV

3. A-322 · Hard · Multiple choice · **No calculator needed**

For real numbers x and y, $x + y < 12$ and $x - y > 4$. Which of the following must be true?

A) $y > 4$

B) $x > 8$

C) $y < 4$

D) $x < 8$

4. A-326 · Hard · SPR · **No calculator needed**

In the xy-plane, the region with $x \geq 0$, $y \geq 0$ and $2x + y \leq 12$ is a triangle. The line $y = mx$ divides this triangle into two regions of equal area. What is the value of m?

5. **A-331** · Hard · Multiple choice · **No calculator needed**

If $7x - 2y = 3x + 6y$ and x and y are nonzero, what is the value of $\frac{x+y}{x-y}$?

A) The value cannot be determined from the information given.

B) $\frac{1}{3}$

C) 2

D) 3

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6. **A-333** · Hard · Multiple choice · **No calculator needed**

A wall is $9x$ meters long, where x is a positive integer. Lamps are placed at both ends of the wall and every $\frac{9}{7}$ meters along it. Which expression gives the number of lamps?

A) $\frac{81x}{7} + 1$

B) $7x$

C) $7x + 1$

D) $9x + 1$

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7. **A-337** · Hard · Multiple choice · **No calculator needed**

The solution to the system $ax + by = 31$ and $bx + ay = 19$, where a and b are constants, is $(4, 1)$. What is the value of $a^2 - b^2$?

A) 16

B) 40

C) 10

D) 58

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8. **A-339** · Hard · Multiple choice · **No calculator needed**

For every real number m, the graph of $y = mx + 3m - 2$ in the xy-plane passes through the point P. What are the coordinates of P?

A) $(-3, -2)$

B) $(0, -2)$

C) $(3, -2)$

D) $(-2, 3)$

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9. **A-342** · Hard · Multiple choice · **No calculator needed**

For how many integer values of k does the point (k, k) satisfy both $y \geq 2x - 5$ and $y > -x + 1$?

A) 5

B) 4

C) Infinitely many

D) 6

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10. **A-324** · Medium · Multiple choice · **No calculator needed**

A farm has 40 hectares of date palms and 25 hectares of olive trees. The equation $40x + 25y = 3{,}600$ represents the farm's total harvest, in tonnes, last season. What does x represent?

A) The average date harvest per hectare, in tonnes

B) The number of tonnes harvested per date palm tree

C) The number of hectares of date palms

D) The total date harvest, in tonnes

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11. **A-327** · Medium · Multiple choice · **No calculator needed**

The function $V(t) = 1{,}200 - 0.75t$ gives the volume, in liters, of water in a tank t minutes after it starts draining. Which function gives the volume, in liters, h hours after the tank starts draining?

A) $1{,}200 - 0.75(h + 60)$

B) $1{,}200 - 45h$

C) $20 - 0.75h$

D) $1{,}200 - 0.0125h$

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12. **A-330** · Medium · SPR · **No calculator needed**

A shopper spends exactly 600 dollars on x kilograms of oranges at 4 dollars per kilogram and y kilograms of apples at 5 dollars per kilogram, so $4x + 5y = 600$. If the shopper buys 10 more kilograms of oranges and still spends 600 dollars, by how many kilograms does y decrease?

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13. **A-341** · Medium · Multiple choice · **No calculator needed**

Which inequality has the solution set $x \leq -3$ or $x \geq 11$?

A) $|x + 4| \geq 7$

B) $|x - 4| \leq 7$

C) $|x - 4| \geq 7$

D) $|x - 7| \geq 4$

Advanced Math (14 items)

14. **AM-310** · Hard · Multiple choice · **Desmos**

How many real solutions does the equation $x^2 = 2^x$ have?

A) 1

B) 2

C) 3

D) 4

15. **AM-326** · Hard · Multiple choice · **Desmos**

What is the sum of all distinct real solutions of $x^3 - 2x^2 - 15x + 36 = 0$?

A) 2

B) 7

C) -2

D) -1

16. **AM-304** · Hard · SPR · **No calculator needed**

The function g is defined by $g(x) = \dfrac{x^2 - kx - 18}{x + 3}$, where k is a constant. The graph of $y = g(x)$ in the xy-plane is a line with exactly one point removed. What is the y-coordinate of the removed point?

17. **AM-306** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = x^2 - 9$ for $x < 2$ and $h(x) = -2x + 10$ for $x \geq 2$. What is the sum of all values of m for which $h(m) = 0$?

A) 0

B) 2

C) 5

D) 8

18. **AM-308** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = x^2 - 1$. How many distinct real solutions does the equation $f(f(x)) = 0$ have?

A) 1

B) 4

C) 2

D) 3

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19. **AM-313** · Hard · Multiple choice · **No calculator needed**

A kiosk's daily profit, in hundreds of dollars, is modeled by $P(x) = -2x^2 + 12x + 5$, where x is the price of an item in dollars. The model is used only for $4 \le x \le 7$. What is the greatest daily profit, in hundreds of dollars, predicted by the model on this interval?

A) 21

B) 23

C) -9

D) 5

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20. **AM-315** · Hard · Multiple choice · **No calculator needed**

When the polynomial $p(x)$ is divided by $x - 5$, the remainder is 8. What is the remainder when $p(2x - 1)$ is divided by $x - 3$?

A) 15

B) 8

C) 5

D) 3

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21. **AM-319** · Hard · SPR · **No calculator needed**

The function f is defined by $f(x) = a\big((x-3)^2 - b\big)\big((x-3)^2 - c\big)$, where a, b and c are constants. If $f(1) = 40$ and $f(6) = -14$, what is the value of $f(5) + f(0)$?

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22. **AM-321** · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{x^2}{\sqrt{x^2 - k^2}} = \dfrac{k^2}{\sqrt{x^2 - k^2}} + 24$, k is a positive constant. Which of the following is a solution?

A) $-\sqrt{k^2 + 576}$

B) $\sqrt{576 - k^2}$

C) $\sqrt{k^2 + 24}$

D) $k + 24$

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23. **AM-323** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 3x^2 - 5x + 2$. Which expression is equivalent to $f(a + 1) - f(a)$?

A) 0

B) $6a - 2$

C) $6a + 3$

D) $6a + 8$

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24. **AM-325** · Hard · Multiple choice · **No calculator needed**

The equations $x^2 - x - 12 = 0$ and $x^2 + kx + 8 = 0$ have a common solution, where k is a positive constant. What is the value of k?

A) -6

B) $\frac{17}{3}$

C) $-\frac{17}{3}$

D) 6

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25. **AM-328** · Hard · Multiple choice · **No calculator needed**

The functions f and g are defined by $f(x) = 7 \cdot 4^x$ and $g(x) = f(x - 1.5)$. For all values of x, $g(x) = k \cdot f(x)$, where k is a constant. What is the value of k?

A) -6

B) $\frac{7}{8}$

C) $\frac{1}{8}$

D) 8

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26. **AM-330** · Hard · Multiple choice · **No calculator needed**

The function g is defined by $g(x) = 3^x$. If $g(a) = 4$, what is the value of $g(2a - 1)$?

A) $\frac{16}{3}$

B) 15

C) $\frac{8}{3}$

D) 7

27. **AM-318** · Medium · SPR · **No calculator needed**

How many integers x satisfy the inequality $x(x-4) < 2x$?

Geometry and Trigonometry (22 items)

28. **GT-253** · Hard · Multiple choice · **Calculator**

Square pyramids X and Y have the same base, a square with side length 10. Pyramid X has height 12 and pyramid Y has height 24. The total surface area of Y is approximately what percent greater than the total surface area of X?

A) 89%

B) 100%

C) 64%

D) 50%

29. **GT-250** · Hard · SPR · **Desmos**

In the equation $\sin\left(\frac{39\pi}{p}\right) = \frac{\sqrt{3}}{2}$, p is a constant and $p > 60$. What is the value of p?

30. **GT-233** · Hard · SPR · **No calculator needed**

The graph of $x^2 + y^2 + 10x - 8y + k = 0$ in the xy-plane is a circle, where k is a constant. What is the greatest possible integer value of k?

31. **GT-235** · Hard · Multiple choice · **No calculator needed**

Points A, B and C lie on a circle with radius 12, and C lies on the major arc AB. The measure of angle ACB is $\frac{2\pi}{9}$ radians. What is the length of minor arc AB?

A) $\frac{16\pi}{3}$

B) $\frac{8\pi}{3}$

C) $\frac{64\pi}{3}$

D) $\frac{56\pi}{3}$

32. GT-237 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle, $AC = 16$ and $BC = 12$. Point E lies on $\overline{AC}$ with $AE = 10$, and point F lies on $\overline{AB}$ so that $\overline{EF} \perp \overline{AB}$. What is the length of $\overline{EF}$?

A) 8

B) 6

C) 7.5

D) 12.5

33. GT-240 · Hard · Multiple choice · **No calculator needed**

A right circular cone stands with its vertex pointing up. Its base radius is 6 and its height is 12. Water fills the cone to a depth of 6, measured from the base. What is the volume of the water?

A) 18π

B) 72π

C) 108π

D) 126π

34. GT-242 · Hard · Multiple choice · **No calculator needed**

Line ℓ passes through $(0, 0)$ and $(a, 12)$, where $a > 0$. The acute angle between ℓ and the positive y-axis measures $60°$. What is the value of a?

A) $12\sqrt{3}$

B) $4\sqrt{3}$

C) 24

D) 6

35. GT-244 · Hard · Multiple choice · **No calculator needed**

A square has one side on the diameter of a semicircle of radius 10, and its other two vertices lie on the arc. What is the area of the square?

A) 100

B) 50

C) 80

D) 200

36. GT-247 · Hard · Multiple choice · **No calculator needed**

In a circle of radius 13, two parallel chords have lengths 24 and 10. Which of the following could be the distance between the two chords?

A) 12

B) 17

C) 26

D) 5

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37. GT-251 · Hard · Multiple choice · **No calculator needed**

A rectangle is inscribed in a circle with area 676π. The length of the rectangle's diagonal is twice the length of its shorter side. What is the perimeter of the rectangle?

A) $52 + 52\sqrt{3}$

B) 104

C) $26 + 26\sqrt{3}$

D) $104 + 52\sqrt{3}$

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38. GT-254 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $18°$. Which additional fact is NOT sufficient to determine the measure of angle B?

A) $m\angle B = m\angle C$

B) $m\angle C = 3\,m\angle B$

C) Triangle ABC is isosceles.

D) $m\angle C = 90°$

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39. GT-257 · Hard · Multiple choice · **No calculator needed**

In triangle JKL, the measure of angle J is $(ab)°$ and the measure of angle K is $(cd)°$, where $ab + cd = 90$. If $\sin J = 0.6$, what is the value of $\tan K$?

A) $\frac{3}{4}$

B) $\frac{4}{3}$

C) $\frac{4}{5}$

D) The value cannot be determined from the information given.

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40. GT-259 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, $AB = AC = 17$ and $BC = 16$. What is the value of $\tan B$?

A) $\frac{8}{15}$

B) $\frac{15}{8}$

C) $\frac{15}{17}$

D) $\frac{16}{17}$

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41. GT-261 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle and $\cos A = k$. Which expression is equal to $\sin^2 A + \sin^2 B$?

A) 1

B) The value cannot be determined from the information given.

C) $2 - 2k^2$

D) $2k^2$

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42. GT-263 · Hard · Multiple choice · **No calculator needed**

A straight railway track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 300 meters long, measured along the slope. What is the vertical rise of the track, in meters?

A) 400

B) 240

C) 180

D) 225

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43. GT-265 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle B is $45°$ and the measure of angle C is $30°$. The altitude from A to $\overline{BC}$ has length 6. What is the length of $\overline{BC}$?

A) 12

B) $6 + 2\sqrt{3}$

C) $12 + 6\sqrt{2}$

D) $6 + 6\sqrt{3}$

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44. GT-268 · Hard · Multiple choice · **No calculator needed**

From the top of a cliff 45 meters high, the angles of depression to two boats on opposite sides of the cliff are $30°$ and $45°$. The boats and the base of the cliff are on a straight line. What is the distance, in meters, between the boats?

A) $45\sqrt{3} - 45$

B) $45 + 15\sqrt{3}$

C) $45\sqrt{3}$

D) $45 + 45\sqrt{3}$

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45. GT-270 · Hard · SPR · **No calculator needed**

A rectangle has side lengths 6 and $6\sqrt{3}$. What is the measure, in degrees, of the acute angle formed by its diagonals?

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46. GT-272 · Hard · Multiple choice · **No calculator needed**

Parallel lines ℓ and m are horizontal, with ℓ above m. Point P lies between the lines. Point A on ℓ and point B on m are both to the right of P. Segment $\overline{PA}$ makes a $38°$ angle with line ℓ, and segment $\overline{PB}$ makes a $47°$ angle with line m, both angles opening toward P. What is the measure, in degrees, of angle APB?

A) 42.5

B) 95

C) 85

D) 9

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47. GT-248 · Medium · Multiple choice · **Calculator**

The radius of a right circular cylinder is increased by 20% and its height is decreased by 25%. What is the effect on the volume of the cylinder?

A) It does not change.

B) It increases by 8%.

C) It decreases by 5%.

D) It increases by 19%.

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48. GT-245 · Medium · Multiple choice · **No calculator needed**

In the xy-plane, the endpoints of a diameter of a circle are $(-2, 3)$ and $(6, -1)$. Which equation represents the circle?

A) $x^2 + y^2 - 4x - 2y - 15 = 0$

B) $x^2 + y^2 - 4x - 2y - 20 = 0$

C) $x^2 + y^2 - 4x - 2y - 75 = 0$

D) $x^2 + y^2 + 4x + 2y - 15 = 0$

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49. GT-274 · Medium · Multiple choice · **No calculator needed**

In triangle ABC, $AB = 2x + 1$, $BC = x + 6$ and $CA = 3x - 4$. The perimeter of the triangle is 45. Which angle of the triangle has the greatest measure?

A) Angle A

B) It cannot be determined from the information given.

C) Angle C

D) Angle B

Problem-Solving and Data Analysis (27 items)

50. PS-166 · Hard · Multiple choice · **Calculator**

District A has $120{,}000$ residents and a population density of $4{,}000$ residents per square kilometer. District B has $60{,}000$ residents and a population density of $1{,}500$ residents per square kilometer. Which is closest to the population density, in residents per square kilometer, of the two districts combined?

A) $5{,}500$

B) $3{,}167$

C) $2{,}571$

D) $2{,}750$

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51. PS-178 · Hard · Multiple choice · **Calculator**

A store's profit on each jacket is 37.5% of the selling price. The profit is what percent of the cost of the jacket to the store?

A) 37.5%

B) 27.3%

C) 62.5%

D) 60%

52. PS-181 · Hard · Multiple choice · **Calculator**

A worker spends 30% of her monthly salary on rent, then 25% of the remaining amount on food, then 40% of what is left after that on savings. She has $3{,}780$ dollars left. What is her monthly salary, in dollars?

A) $75{,}600$

B) $7{,}200$

C) $12{,}000$

D) $6{,}300$

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53. PS-192 · Hard · Multiple choice · **Calculator**

A bank sells US dollars at 3.77 riyals per dollar. It first deducts a 1% commission from the riyals a customer pays. How many US dollars does a customer receive for $5{,}655$ riyals?

A) $1{,}515$

B) $1{,}485$

C) $21{,}319$

D) $1{,}500$

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54. PS-159 · Hard · SPR · **No calculator needed**

The data set $12, 15, 19, 24, 30$ has a range of 18. A sixth value, x, is added, and the range of the new data set is 1.5 times the original range. What is the sum of all possible values of x?

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55. PS-161 · Hard · SPR · **No calculator needed**

A data set contains the values $1, 2, 3, 4, 5$ with frequencies $4, k, 6, 3, 5$, respectively, where k is a positive integer. The median of the data set is 3. What is the greatest possible value of k?

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56. PS-163 · Hard · Multiple choice · **No calculator needed**

Data set A has 11 values with median 40. Data set B has 11 values with median 52. The two data sets are combined. Which of the following could be the median of the combined 22 values?

A) 46

B) 38

C) 60

D) 54

57. **PS-168** · Hard · SPR · **No calculator needed**

In a school, a student is chosen at random. The probability that the student is in grade 10, given that the student is in the robotics club, is $\frac{3}{8}$. The probability that the student is in the robotics club, given that the student is in grade 10, is $\frac{1}{4}$. There are 96 grade 10 students. How many robotics club members are NOT in grade 10?

58. **PS-170** · Hard · Multiple choice · **No calculator needed**

In grade 10, 18 students are in the robotics club and 42 are not. In grade 11, 30 students are in the robotics club and 60 are not. A student is chosen at random from the students who are in grade 10 OR in the robotics club. What is the probability that the student is in grade 11?

A) $\frac{1}{3}$

B) $\frac{1}{5}$

C) $\frac{1}{2}$

D) $\frac{5}{8}$

59. **PS-172** · Hard · SPR · **No calculator needed**

The model $\hat{y} = 4.8x + 62$ estimates a town's population y, in thousands, where x is the number of decades since 1980. According to the model, what is the estimated increase in the population, in people, per year?

60. **PS-175** · Hard · SPR · **No calculator needed**

Data set A has 10 integers: 4 are from 10 to 19 and 6 are from 20 to 29. Data set B has 10 integers: 3 are from 30 to 39 and 7 are from 40 to 49. What is the smallest possible value of (mean of B) $-$ (mean of A)?

61. **PS-180** · Hard · Multiple choice · **No calculator needed**

A club has 40 members, and 30% of the members are girls. How many girls must join the club, with no one leaving, so that girls make up 60% of the members?

A) 20

B) 12

C) 30

D) 28

62. **PS-184** · Hard · Multiple choice · **No calculator needed**

Fresh dates are 80% water by mass. After drying, the dates are 50% water by mass. How many kilograms of dried dates are produced from 60 kilograms of fresh dates?

A) 30

B) 24

C) 42

D) 12

63. **PS-186** · Hard · Multiple choice · **No calculator needed**

At a school, 85% of the boys and 60% of the girls passed an exam, and 70% of all the students passed. What percent of the students at the school are boys?

A) 25%

B) 50%

C) 60%

D) 40%

64. **PS-189** · Hard · Multiple choice · **No calculator needed**

A crew of 12 workers can finish a job in 20 days. After the crew has worked for 5 days, 3 more workers join, all working at the same rate. How many days in total does the job take?

A) 16

B) 25

C) 17

D) 12

65. **PS-191** · Hard · Multiple choice · **No calculator needed**

A printer prints p pages every m seconds. Which expression gives the number of minutes the printer takes to print n pages?

A) $\dfrac{60m}{pn}$

B) $\dfrac{60mn}{p}$

C) $\dfrac{pn}{60m}$

D) $\dfrac{mn}{60p}$

66. **PS-194** · Hard · Multiple choice · **No calculator needed**

A data set consists of $12, 15, 18, 21, 24, 27, 30, 33, 36, 40$. The value 40 is mistakenly recorded as 400. How do the median, the interquartile range (IQR) and the standard deviation of the recorded data compare with those of the correct data?

A) Only the median is unchanged.

B) The median, IQR and standard deviation are all unchanged.

C) The median is unchanged, but the IQR and standard deviation are greater.

D) The median and IQR are unchanged, but the standard deviation is greater.

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67. **PS-196** · Hard · Multiple choice · **No calculator needed**

At a company, the median monthly salary is $9{,}000$ dollars and the mean monthly salary is $14{,}500$ dollars. Which statement must be true?

A) The most common salary is $9{,}000$ dollars.

B) The standard deviation of the salaries is $5{,}500$ dollars.

C) No more than half of the employees earn more than $14{,}500$ dollars.

D) More than half of the employees earn more than $14{,}500$ dollars.

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68. **PS-198** · Hard · Multiple choice · **No calculator needed**

A data set has six values. Their deviations from 40 (value minus 40) are -7, 3, 12, -5, x and 2. The mean of the data set is 42. What is the value whose deviation from 40 is x?

A) 35

B) 7

C) 47

D) 52

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69. **PS-201** · Hard · Multiple choice · **No calculator needed**

The recorded values of a data set are $18, 22, 27, 30, 33, 36, 41, 47, 52$. The value 52 should have been 25. After the correction, how do the mean and the median change?

A) The mean decreases by 3 and the median is unchanged.

B) The median decreases and the mean is unchanged.

C) The mean decreases by 27 and the median decreases by 3.

D) The mean and the median both decrease by 3.

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70. **PS-203** · Hard · Multiple choice · **No calculator needed**

In an experiment, garden plots were randomly assigned to receive 0, 20 or 40 grams of a fertilizer. The plots given 40 grams had the highest mean yield. Which conclusion is best supported?

A) Doubling the fertilizer doubles the yield.

B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

C) No causal conclusion is possible.

D) Using 80 grams would give an even higher yield.

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71. **PS-209** · Hard · Multiple choice · **No calculator needed**

For data set P, the line of best fit is $\hat{y} = 4.2x + 3$ and the correlation coefficient is 0.41. For data set Q, the line of best fit is $\hat{y} = 0.6x + 10$ and the correlation coefficient is 0.93. Which statement is true?

A) The associations cannot be compared because the slopes are different.

B) P has the stronger linear association because its slope is greater.

C) Q has a weaker association because its slope is less than 1.

D) Q has the stronger linear association.

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72. **PS-177** · Medium · Multiple choice · **Calculator**

The monthly rent of an apartment increased by 60%. By what percent must the new rent decrease to return to the original rent?

A) 62.5%

B) 37.5%

C) 60%

D) 40%

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73. **PS-158** · Medium · SPR · **No calculator needed**

The daily high temperatures in a city for one week have a mean of $34°$C and a standard deviation of $4°$C. Each temperature is converted to degrees Fahrenheit using $F = 1.8C + 32$. What is the standard deviation of the converted temperatures, in degrees Fahrenheit?

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74. **PS-200** · Medium · SPR · **No calculator needed**

In a data set, a value is called a low outlier if it is less than $Q_1 - 1.5 \cdot \text{IQR}$, where Q_1 is the first quartile and IQR is the interquartile range. For a data set, $Q_1 = 28$ and the third quartile is 40. What is the greatest integer value that would be a low outlier?

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75. **PS-206** · Medium · Multiple choice · **No calculator needed**

To estimate the average weekly exercise time of adults in a city, a researcher randomly selected 400 people from the membership lists of the city's gyms. Why is the estimate likely to be biased?

A) The sample size is too small.

B) The selection was random, so the estimate is not biased.

C) The list leaves out adults who are not gym members.

D) There was no random assignment.

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76. **PS-208** · Medium · Multiple choice · **No calculator needed**

For a data set, the correlation coefficient between height in centimeters and mass in kilograms is $r = 0.72$. The heights are converted to inches, the masses to pounds, and the two variables are swapped between the axes. What is the correlation coefficient for the converted data?

A) 0.72

B) -0.72

C) $0.72 \cdot \dfrac{2.2}{2.54}$

D) $\dfrac{1}{0.72}$
